

AP Calculus BC

Course Outline • Mathlify Summer Program • College Calculus I & II Preparation

Course Overview

AP Calculus BC is equivalent to College Calculus I and II combined. The course develops understanding of limits, derivatives, integrals, and series through graphical, numerical, analytical, and verbal representations. Students are prepared to enter the school year confident, organized, and ready to succeed on the AP exam.

Course Units

Unit 1 — Limits & Continuity

- Limits from graphs, tables, and algebra
- One-sided limits and limits at infinity
- Continuity and Intermediate Value Theorem
- Asymptotic and end behavior of functions

Unit 2 — Derivatives: Definition & Rules

- Derivative as a limit of the difference quotient
- Power, product, quotient, and chain rules
- Trig, inverse trig, exponential, and logarithmic derivatives
- Implicit, parametric, and polar differentiation

Unit 3 — Applications of Derivatives

- Tangent lines and rates of change
- Critical points, extrema, and concavity
- First and second derivative tests
- Optimization and related rates problems
- Mean Value Theorem and Rolle's Theorem

Unit 4 — Integrals & Accumulation

- Riemann sums and definite integrals
- Area under and between curves
- Fundamental Theorem of Calculus
- Trapezoidal Rule and accumulation functions

Unit 5 — Integration Techniques & Inverse Functions

- Antiderivatives and initial value problems
- Substitution and integration by parts
- Partial fractions
- Inverse trig and logarithmic integrals
- Improper integrals

Unit 6 — Differential Equations & Slope Fields

- Slope fields and solution curves
- Euler's method
- Separable differential equations
- Exponential and logistic growth models

Unit 7 — Advanced Applications of Integrals

- Volumes by slicing
- Disk, washer, and shell methods
- Volumes with known cross sections
- Arc length and motion along a line

Unit 8 — BC-Only Topics: Improper Integrals & Series

- Improper integrals and convergence
- L'Hôpital's Rule
- Sequences and series
- Geometric, p-series, and alternating series
- Taylor and Maclaurin polynomials
- Error bounds and approximation

Assessment & Practice

Students complete regular AP-style multiple choice and free-response questions. Emphasis is placed on clear written justification, multiple representations, and strategic calculator usage.